The well:
five faces of the ground state of the windowed Weil form, and the approximation lemma that would hold them

D. Joubert (with the assistance of two language models)

6 September 2026, v3 (the law) — consolidates notebook §16, §66–68, §86–89, §120–135 and Grok's `lemma-theta-v-consolidated.md` of `riemann-weil-phenomenology`

Abstract

On the window $[0, L]$, $L = \log \mu$, the truncated Weil form Q_L (archimedean part minus the prime towers up to μ) is positive with a smallest eigenvalue λ_0 that falls super-exponentially with μ — the *well*. Its ground state v has been measured from two sides over a week, by two independent chains (one on the space side, one on the zero side), on ζ, fifteen Dirichlet characters, eight elliptic curves and two Dedekind products, and the measurements fit one object: v is the unit-norm function of the window whose Fourier transform is hyper-null at the in-band zeros, whose spectral mass sits in the desert below the first zero, whose autocorrelation collapses super-exponentially at every lag beyond $y \approx 1.2$ (this is the "silence at the primes" of the quorum mechanism, which is not about primes), and whose value at the window edge is what remains: $\lambda_0 \approx \psi(0)^2 \cdot S$, with S the leakage of an edge jump onto the zeros beyond the band edge, computable from the zeros alone — the edge-doubling law $-\ln \lambda_0 = 2(-\ln |\psi(0)|) + O(1)$ found by Grok and verified on thirteen windows (the edge carries 82–103% of the depth). The depth law that holds them is a count: the eigenvalues of the zero Gram form a plunge ladder with near-universal rungs (16 → 5 nats, mean $\bar{r} \approx 11$) and the number of exponentially small eigenvalues equals $D_{\max} = \max_\gamma(\gamma L/2\pi - N_\Gamma(\gamma))$, the maximal lead of the window's Nyquist count over the zero count (a discrete Landau theorem, checked $10/10, 5/5, 3/3, 1/1$); hence $\ell \approx 11.0\, D_{\max}$, within 3% on eight of twelve degree-1 windows and 10% on GL_2, with $D_{\max}$ read off the zero list and no fitted parameter but the mean rung. The marginal weight of each in-band zero is $w(\gamma) \approx w_0(1 - \gamma/\gamma_c)$ with $w_0 \approx 11$ universal and γ_c the Nyquist crossing (fourteen windows). The note states the object, the evidence, the dead ends, the law, and what remains: the proof of the discrete count and the origin of the constant 11. Nothing here bears on RH.

1 The object

Q_L on the hat/cosine basis $\{\eta_n\}_{n \le N}$ of $[0, L]$; $\lambda_0 = \min Q_L$, v its eigenvector, $\psi = \sum v_n \eta_n$ the function, $\hat{v}$ its Fourier transform, $\Theta_v(y) = \int \psi(t)\psi(t+y)\, dt$ its autocorrelation, $\ell = -\ln \lambda_0$ the depth, $s = \ell/\mu$ the drilling speed, $\omega_{\max} = 2\pi N/L$ the band edge. Under RH, $Q_L(f) = \sum_\gamma |\hat{f}(\gamma)|^2$ over the zeros; the repository's identity $Q_L =$ zero Gram holds to the tail on every L-function tried (`zeros-from-the-radical`, `gl2-prime-side`).

2 Five faces

1. The desert (§66–68, sampling-floor). λ_0 is the sampling constant of the zero set on the window, made small by the desert $[0, \gamma_1)$ where there are no zeros: under RH $c_L > 0$ (Theorem 1 of `sampling-floor`) and $\lambda_0(N) \downarrow c_L$. The Slepian cost of the desert alone, $1.71\, L(\gamma_1 - \nu)_+/\mu$ per unit μ, is 60–90% of s on the fast characters and on the rank-0 elliptic curves.

2. Hyper-nullity and the edge budget (§16.3–16.7). $|\hat{v}(\gamma_k)|$ at the in-band zeros is exponentially smaller than any bound requires (4.7×10^{-72} at γ_1 for ζ, $\mu=16$) — the extremity law $|\hat{v}(\gamma_1)|\approx C\lambda_0$; the budget $\lambda_0=\sum_k|\hat{v}(\gamma_k)|^2$ is carried by the zeros at and beyond the band edge (leakage rate $\tau\approx 0.48$, §16.4).

3. The mass in the desert (§122). $|\hat{v}(\gamma)|^2$ is 0.8 at $\gamma=1$, 0.25 at 5, 5×10^{-3} at 10, then $e^{-1.34\gamma}$ with hyper-null dips at the zeros, 10^{-60} at $\omega_{\max}$ (ζ, $\mu=16$): the norm lives below γ_1. Budget at the edge, mass in the desert — both true, distinct.

4. The collapse of the autocorrelation (§120–123; Grok's space-side lemma). $\Theta_v(y)>0$ everywhere, smooth (no band-edge oscillation at step 0.004), and $-\ln\Theta_v(y)$ grows super-exponentially: $\approx 0.2\,s\,y\,e^y$ on ζ at $\mu=16$ ($R^2=0.9995$, slope ratio $2.9=(1+y)e^y$, Gaussian excluded), with a coefficient $\propto 1/L$ across μ ($a\cdot L=0.62, 0.62, 0.58$ at $\mu=8, 11, 16$) and a shape that the window edge bends: steeper near L, gentler for composite objects. The space side (Grok) reads the same: ψ has a Gaussian bulk, $\psi(t)\approx\psi_{\rm mid}e^{-a(t-L/2)^2}$ with $aL^2=\ell$ on ζ, and steep edges (e^{-s}); the $y\,e^y$ tail of Θ_v is the edge, not the desert weight. The towers sample this collapse at the prime lags: $-\ln\delta_p\approx 0.19\,s\,p\log p$ (one $\Gamma_{\mathbb{R}}$) or $0.38\,s\,p\log p$ (several; degrees 2 and 3 alike, §118–119) is the same law read at $n=p^k$, and at lags with no prime the same numbers appear (§120–121). Consequence for the quorum mechanism: its first half (the ground state is deaf to each prime) is a generic property of the well; only the coupling κ_p, Gaussian in $(p\log p)^2/(\mu\log\mu)$, is arithmetic. The ground state's sum rule sees only 2 and 3 ($98.9\%+1.1\%$ for ζ).

5. The edge value (Grok; §125, §127). $-\ln\lambda_0=2(-\ln|\psi(0)|)+R$, $\psi(0)=L^{-1/2}(v_0+\sqrt{2}\sum_{n\geq1}v_n)$: ratio 2.06–2.19 on ζ, χ_3, χ_{29}, $\zeta L(\chi_3)$; $-2\ln|\psi(0)|/\ell\in[0.82, 0.98]$ on Grok's eleven windows and 0.97–1.03 on ζ — thirteen windows, five orders of magnitude in λ_0, the edge carries the depth. And why: a function with value $\psi(0)$ at both ends has a jump whose transform is $2\psi(0)\sin(\gamma L/2)/\gamma$; in band the smooth part cancels it (hyper-nullity), beyond the band it does not. On ζ at $\mu=11$ with 500 zeros, the zeros beyond $\omega_{\max}$ carry 93.3% of λ_0 and the estimated tail beyond 811 completes it to 1.5%; the jump prediction $\sum_{\gamma_k>\omega_{\max}}8\psi(0)^2\sin^2(\gamma_kL/2)/\gamma_k^2$ matches to a factor 1.55 in the sum, predicting $R=3.37$ against 3.29 measured. Hence

$$\lambda_0\;\approx\;\psi(0)^2\cdot S,\qquad S=\sum_{\gamma_k>\omega_{\max}}\frac{8\sin^2(\gamma_kL/2)}{\gamma_k^2}\times O(1.5)\approx\frac{4}{\pi}\,\frac{\log(\omega_{\max}/2\pi e)}{\omega_{\max}},$$

which explains why R depends on N (through $\omega_{\max}$) and on the object (through the zero density), and why it changes sign when in-band zeros near the band edge still carry budget (ζ, $\mu=16$: $R=-4.4$).

3 One object

The five faces are one statement: *the ground state is the unit-norm function of the window that is hyper-null at the in-band zeros and minimizes its value at the edge; its mass sits in the desert because that is where no constraint holds; its autocorrelation collapses because a smooth bump with steep edges has one; and the depth is the square of the minimal edge value times the leakage of the edge jump onto the zeros beyond the band.* The primes enter twice, and only twice: as the samples of the collapse (harmless), and in the coupling κ_p that interlocks the towers (the arithmetic of the quorum).

4 What is dead

The two-term formula $-\ln c_L \approx aL(\gamma_1 - \nu)_+ + bL\sum(\text{gap} - \nu)_+$ (its gap sum does not converge; at a common cutoff it overpredicts most L-functions, §92–94). The one-set constant by $\lambda_{\max}(\chi_E P \chi_E)$ ($\lambda_{\max}$ sees only the largest interval; Grok, `lemma2-sampling-constant`). The line $\ell \approx a\tau\gamma_1 + b$ (jackknife: the slope halves without the one deep window). The Gaussian form of the collapse in y (slopes). The reading of the silence as arithmetic (§120). Ten pre-registered executions died on this object in one week; the object survived all of them.

5 What is open: the approximation lemma

Lemma 1 (to prove). *Let $\Gamma_L = \{\gamma_k < \omega_{\max}\}$ be the in-band zeros. Among unit-norm functions on $[0, L]$ whose Fourier transform vanishes on Γ_L (to the hyper-null order observed), the minimal edge value satisfies $-\ln|\psi(0)|_{\min} = \frac{1}{2}\, s\,\mu + O(1)$, i.e. $\lambda_0 \asymp \psi(0)^2_{\min}\, S(\omega_{\max})$, with s a functional of the configuration Γ_L: $s = C_0\,\tau\gamma_1/\mu + (\text{gap correction})$, the correction $O(1)$ per cluster of zeros.*

What the data say about s: the desert alone gives a factor-3 dispersion ($\ell/\tau\gamma_1 \in [2.3, 7.4]$ on thirteen windows, ζ the deepest per unit of desert); the gap correction is $O(1)$ per piece in the log-determinant of the one-set compression (Widom's perimeter term: 0.46–0.62 nats per gap, position-independent), which matches ℓ to 20% on χ_5 only; no elementary (γ_1, L) predictor of $|\psi(0)|$ exists. The lemma is pure harmonic analysis — an extremal problem with linear constraints on a window — and it contains no prime. Its proof would give the depth law; its failure would tell which constraint the picture misses.

6 The lemma and the law (§128–135)

The extremal problem. Under the identity, $\lambda_0 = \min_{\|f\|=1} \sum_\gamma |\hat{f}(\gamma)|^2$. In the sub-Nyquist regime (fewer in-band zeros than dimensions) the minimizer annihilates the in-band zeros exactly and pays the out-of-band leakage of its edge jump: the constrained minimum has overlap 1.0000 with the true ground state and 0.62 of its λ_0 (ζ, $\mu = 11$). Above Nyquist (more in-band zeros than dimensions, χ_3, χ_5, all GL_2 windows) exact annihilation is impossible and the well balances hyper-null values instead — which is, to the digit, why ζ drills three times faster than χ_3.

Marginal weights. $w(\gamma) = \ln\lambda_0(G) - \ln\lambda_0(G \setminus \gamma)$, the depth a zero buys. Additive to 2% for local perturbations (the log-determinant is not). Linear in height, $w(\gamma) \approx w_0(1 - \gamma/\gamma_c)$, and *zero above* γ_c: on fourteen windows (server table, `report/marginal-weights.md`) the fitted γ_c is the Nyquist crossing $\gamma L/2\pi = N_\Gamma(\gamma)$ to 10% (88.9/78.1, 64.0/58.4, 52.1/48.0, 31.4/30.7, 19.1/18.6, 10.6/11.5; $\omega_{\max}$ when sub-Nyquist), and $w_0 =$ 10–11.5 nats for every degree-1 window from $N = 41$ to 67 and $\ell = 5.5$ to 143; 8.7–9.4 on 11a1. Zeros denser than the window can resolve are worth nothing; a fictitious zero in the desert is worth 12–18 nats, 3–5 above the line.

The ladder. The eigenvalues of the zero Gram on the window: $-\ln\lambda_k = 107.5, 91.4, 76.9, 63.5, 51.5, 39.9, 29.8,$ then $O(1)$ (ζ, $\mu = 11$, $N = 41$) — exactly ten exponentially small, rungs $16.1, 14.5, 13.4, 12.0, 11.5, 10.2, 9.5, 8.2, 7$ The rungs are the same across objects ($16.1, 16.1, 14.5$ first rungs on ζ, χ_3, χ_5); their *number* is $10, 5, 3, 1$ for $\zeta, \chi_3, \chi_5, \chi_{29}$.

Discrete Landau. That number is $\lceil D_{\max} \rceil$, $D_{\max} = \max_\gamma\big(\gamma L/2\pi - N_\Gamma(\gamma)\big)$: $9.9, 4.96, 3.08, 1.08$. Not $N + 1 - m$ (negative above Nyquist): the maximal lead of the window's Nyquist count over the zero count, because zeros above the crossing free nothing. On GL_2 the same count holds and the central zero of a rank-1 curve counts as one constraint ($37a1$: $D_{\max} - 1 = 0.46$, one small eigenvalue, $\ell = 4.2$).

The depth law. $\ell \approx \bar{r}\, D_{\max}$, $\bar{r} = 11.0$ nats (the mean rung; equal to the w_0 of the marginal law):

window	$D_{\max}$	$11.0\, D_{\max}$	ℓ	
ζ 11 / 16	9.89 / 15.00	108.8 / 165.0	107.0 / 139.7	+1.7% / +18% (sub-N
χ_3 16 / 38	4.96 / 12.32	54.6 / 135.5	55.6 / 139.6	−2% / −
χ_4 16 / 38	3.51 / 9.16	38.6 / 100.8	39.7 / 106.8	−3% / −
χ_5 16 / 38	3.08 / 7.58	33.9 / 83.4	33.8 / 88.3	+0.3% /
χ_8, χ_{13}, χ_{29}, χ_{31}	1.77, 1.19, 1.08, 0.90	19.5, 13.1, 11.9, 9.9	19.8, 10.9, 11.9, 8.5	−1.5%, +20%, 0%,
11a1, 19a1, 67a1 ($\mu = 22$)	2.23, 1.52, 0.67	24.5, 16.7, 7.4	22.2, 15.0, 8.2	±10%

No fitted parameter but $\bar{r}$; $D_{\max}$ is read off the zero list and N. The misses are the ones the structure predicts: $d \approx 1$ (a single rung is not the mean rung) and sub-Nyquist windows where $D_{\max}$ sits at the band edge and grows with N while ℓ saturates. The law replaces the two-term formula, the $\tau\gamma_1$ line and the log-determinant; it contains the desert (where D starts), the gaps (each gap above $2\pi/L$ raises D), the doubled density of GL_2 (small $D_{\max}$) and the rank (-1); it predicts $s = \ell/\mu$ for any object whose zeros are known.

What remains. (A) Prove the discrete Landau count for an arbitrary node set on a finite window. (B) Explain the rung profile $16 \to 5$ and its mean 11 (weakly increasing with N: $10.3 \to 11.5$ from $N = 41$ to 67; smaller in degree 2): the constant of the law. Neither contains a prime. Under RH, (A)+(B) *is* the depth law; without RH the same statements hold for the Gram of whatever the zeros are.

Status

Every number above is measured, with its script and its judge in the repository; the identity $Q_L =$ Gram is the only theorem used and it is used under RH. The two chains — space side (Grok: ψ, $aL^2 = \ell$, edge doubling) and zero side (this note: budget, mass, collapse, leakage) — were run independently and agree. Nothing here bears on RH.